

Role of calcineurin in neurodegeneration produced by misfolded proteins and endoplasmic reticulum stress.

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A hallmark event in neurodegenerative diseases is the accumulation of misfolded aggregated proteins in the brain leading to neuronal dysfunction and disease. Compelling evidence suggests that misfolded proteins damage cells by inducing endoplasmic reticulum (ER) stress and alterations in calcium homeostasis. Changes in cytoplasmic calcium concentration lead to unbalances on several signaling pathways. Recent data suggest that calcium-mediated hyperactivation of calcineurin (CaN), a key phosphatase in the brain, triggers synaptic dysfunction and neuronal death, the two central events responsible for brain degeneration in neurodegenerative diseases. Therefore, blocking CaN hyper-activation might be a promising therapeutic strategy to prevent brain damage in neurodegenerative diseases.

Interaction of Oxidative Stress and Misfolded Proteins in the Mechanism of Neurodegeneration.

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Aggregation of the misfolded proteins β -amyloid, tau, huntingtin, and α -synuclein is one of the most important steps in the pathology underlying a wide spectrum of neurodegenerative disorders, including the two most common ones-Alzheimer's and Parkinson's disease. Activity and toxicity of these proteins depends on the stage and form of aggregates. Excessive production of free radicals, including reactive oxygen species which lead to oxidative stress, is proven to be involved in the mechanism of pathology in most of neurodegenerative disorders. Both reactive oxygen species and misfolded proteins play a physiological role in the brain, and only deregulation in redox state and aggregation of the proteins leads to pathology. Here, we review the role of misfolded proteins in the activation of ROS production from various sources in neurons and glia. We discuss if free radicals can influence structural changes of the key toxic intermediates and describe the putative mechanisms by which oxidative stress and oligomers may cause neuronal death.

Olfactory dysfunction as an early pathogenic indicator in *C. elegans* models of Alzheimer's and polyglutamine diseases.

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Neurodegenerative diseases such as Alzheimer's disease and polyglutamine diseases are characterized by abnormal accumulation of misfolded proteins, leading to neuronal dysfunction and subsequent neuron death. However, there is a lack of studies that integrate molecular, morphological, and functional analyses in neurodegenerative models to fully characterize these time-dependent processes. In this study, we used *C. elegans* models expressing A β 1-42 and polyglutamine to investigate early neuronal pathogenic features in olfactory neurons. Both models demonstrated significant reductions in odor sensitivity in AWB and AWC chemosensory neurons as early as day 1 of adulthood, while AWA chemosensory neurons showed no such decline, suggesting cell-type-specific early neuronal dysfunction. At the molecular level, A β 1-42 or Q40 expression caused age-dependent protein aggregation and

morphological changes in neurons. By day 6, both models displayed prominent protein aggregates in neuronal cell bodies and neurites. Notably, AWB neurons in both models showed significantly shortened cilia and increased instances of enlarged cilia as early as day 1 of adulthood. Furthermore, AWC neurons expressing A β 1-42 displayed calcium signaling defects, with significantly reduced responses to odor stimuli on day 1, further supporting early behavioral dysfunction. In contrast, AWA neuron did not exhibit reduced calcium responses, consistent with the absence of detectable decreases in olfactory sensitivity in these neurons. These findings suggest that decreased calcium signaling and dysfunction in specific sensory neuron subtypes are early indicators of neurodegeneration in *C. elegans*, occurring prior to the formation of visible protein aggregates. We found that the ER unfolded protein response (UPR) is significantly activated in worms expressing A β 1-42. Activation of the AMPK pathway alleviates olfactory defects and reduces fibrillar A β in these worms. This study underscores the use of *C. elegans* olfactory neurons as a model to elucidate mechanisms of proteostasis in neurodegenerative diseases and highlights the importance of integrated approaches.

Small molecules to perform big roles: The search for Parkinson's and Huntington's disease therapeutics.

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Neurological motor disorders (NMDs) such as Parkinson's disease and Huntington's disease are characterized by the accumulation and aggregation of misfolded proteins that trigger cell death of specific neuronal populations in the central nervous system. Differential neuronal loss initiates the impaired motor control and cognitive function in the affected patients. Although major advances have been carried out to understand the molecular basis of these diseases, to date there are no treatments that can prevent, cure, or significantly delay the progression of the disease. In this context, strategies such as gene editing, cellular therapy, among others, have gained attention as they effectively reduce the load of toxic protein aggregates in different models of neurodegeneration. Nevertheless, these strategies are expensive and difficult to deliver into the patients' nervous system. Thus, small molecules and natural products that reduce protein aggregation levels are highly sought after. Numerous drug discovery efforts have analyzed large libraries of synthetic compounds for the treatment of different NMDs, with a few candidates reaching clinical trials. Moreover, the recognition of new druggable targets for NMDs has allowed the discovery of new small molecules that have demonstrated their efficacy in pre-clinical studies. It is also important to recognize the contribution of natural products to the discovery of new candidates that can prevent or cure NMDs. Additionally, the repurposing of drugs for the treatment of NMDs has gained huge attention as they have already been through clinical trials confirming their safety in humans, which can accelerate the development of new treatment. In this review, we will focus on the new advances in the discovery of small molecules for the treatment of Parkinson's and Huntington's disease. We will begin by discussing the available pharmacological treatments to modulate the progression of neurodegeneration and to alleviate the motor symptoms in these diseases. Then, we will analyze those small molecules that have reached or are currently under clinical trials, including natural products and repurposed drugs.

TRIM family proteins: roles in proteostasis and neurodegenerative diseases.

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Neurodegenerative diseases (NDs) are a diverse group of disorders characterized by the progressive degeneration of the structure and function of the central or peripheral nervous systems. One of the major features of NDs, such as Alzheimer's disease (AD), Parkinson's disease (PD) and Huntington's disease (HD), is the aggregation of specific misfolded proteins, which induces cellular dysfunction, neuronal death, loss of synaptic connections and eventually brain damage. By far, a great amount of evidence has suggested that TRIM family proteins play crucial roles in the turnover of normal regulatory and misfolded proteins. To maintain cellular protein quality control, cells rely on two major classes of proteostasis: molecular chaperones and the degradative systems, the latter includes the ubiquitin-proteasome system (UPS) and autophagy; and their dysfunction has been established to result in various physiological disorders including NDs. Emerging evidence has shown that TRIM proteins are key players in facilitating the clearance of misfolded protein aggregates associated with neurodegenerative disorders. Understanding the different pathways these TRIM proteins employ during episodes of neurodegenerative disorder represents a promising therapeutic target. In this review, we elucidated and summarized the diverse roles with underlying mechanisms of members of the TRIM family proteins in NDs.

The Future of Seed Amplification Assays and Clinical Trials.

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Prion-like seeded misfolding of host proteins is the leading hypothesised cause of neurodegenerative diseases. The exploitation of the mechanism in the protein misfolding cyclic amplification (PMCA) and real-time quaking-induced conversion (RT-QuIC) assays have transformed prion disease research and diagnosis and have steadily become more widely used for research into other neurodegenerative disorders. Clinical trials in adult neurodegenerative diseases have been expensive, slow, and disappointing in terms of clinical benefits. There are various possible factors contributing to the failure to identify disease-modifying treatments for adult neurodegenerative diseases, some of which include: limited accuracy of antemortem clinical diagnosis resulting in the inclusion of patients with the "incorrect" pathology for the therapeutic; the role of co-pathologies in neurodegeneration rendering treatments targeting one pathology alone ineffective; treatment of the primary neurodegenerative process too late, after irreversible secondary processes of neurodegeneration have become established or neuronal loss is already extensive; and preclinical models used to develop treatments not accurately representing human disease. The use of seed amplification assays in clinical trials offers an opportunity to tackle these problems by sensitively detecting in vivo the proteopathic seeds thought to be central to the biology of neurodegenerative diseases, enabling improved diagnostic accuracy of the main pathology and co-pathologies, and very early intervention, particularly in patients at risk of monogenic forms of neurodegeneration. The possibility of quantifying proteopathic seed load, and its reduction by treatments, is an attractive pharmacodynamic biomarker in the preclinical and early clinical stages of drug development. Here we review some potential applications of seed amplification assays in clinical trials.

Protein folding alterations in amyotrophic lateral sclerosis.

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Protein misfolding leads to the formation of aggregated proteins and protein inclusions, which are associated with synaptic loss and neuronal death in neurodegenerative diseases. Amyotrophic lateral

sclerosis (ALS) is a neurodegenerative disease that targets motor neurons in the brain, brainstem and spinal cord. Several proteins misfold and are associated either genetically or pathologically in ALS, including superoxide dismutase 1 (SOD1), Tar DNA binding protein-43 (TDP-43), Ubiquilin-2, p62, VCP, and dipeptide repeat proteins produced by unconventional repeat associated non-ATG translation of the repeat expansion in C9ORF72. Chaperone proteins, including heat shock proteins (Hsp's) and the protein disulphide isomerase (PDI) family, assist in protein folding and therefore can prevent protein misfolding, and have been implicated as being protective in ALS. In this review we provide an overview of the current literature regarding the molecular mechanisms of protein misfolding and aggregation in ALS, and the role of chaperones as potential targets for therapeutic intervention. This article is part of a Special Issue entitled SI:ER stress.

The therapeutic potential of chemical chaperones in protein folding diseases.

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Several neurodegenerative diseases are caused by defects in protein folding, including Alzheimer, Parkinson, Huntington, and prion diseases. Once a disease-specific protein misfolds, it can then form toxic aggregates which accumulate in the brain, leading to neuronal dysfunction, cell death, and clinical symptoms. Although significant advances have been made toward understanding the mechanisms of protein aggregation, there are no curative treatments for any of these diseases. Since protein misfolding and the accumulation of aggregates are the most upstream events in the pathological cascade, rescuing or stabilizing the native conformations of proteins is an obvious therapeutic strategy. In recent years, small molecules known as chaperones have been shown to be effective in reducing levels of misfolded proteins, thus minimizing the accumulation of aggregates and their downstream pathological consequences. Chaperones are classified as molecular, pharmacological, or chemical. In this mini-review we summarize the modes of action of different chemical chaperones and discuss evidence for their efficacy in the treatment of protein folding diseases in vitro and in vivo.

Targeting chaperones, heat shock factor-1, and unfolded protein response: Promising therapeutic approaches for neurodegenerative disorders.

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Protein misfolding, which is known to cause several serious diseases, is an emerging field that addresses multiple therapeutic areas. Misfolding of a disease-specific protein in the central nervous system ultimately results in the formation of toxic aggregates that may accumulate in the brain, leading to neuronal cell death and dysfunction, and associated clinical manifestations. A large number of neurodegenerative diseases in humans, including Alzheimer's, Parkinson's, Huntington's, and prion diseases, are primarily caused by protein misfolding and aggregation. Notably, the cellular system is equipped with a protein quality control system encompassing chaperones, ubiquitin proteasome system, and autophagy, as a defense mechanism that monitors protein folding and eliminates inappropriately folded proteins. As the intrinsic molecular mechanisms of protein misfolding become more clearly understood, the novel therapeutic approaches in this arena are gaining considerable interest. The present review will describe the chaperones network and different approaches as the therapeutic targets for neurodegenerative diseases. Current and emerging therapeutic approaches to combat

neurodegenerative diseases, addressing the roles of molecular, chemical, and pharmacological chaperones, as well as heat shock factor-1 and the unfolded protein response, are also discussed in detail.

[Keywords for study of pathological conditions of neurodegenerative diseases].

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Molecular genetic analyses revealed that most neurodegenerative diseases including Alzheimer's disease, Parkinson's disease, amyotrophic lateral sclerosis (ALS), and the polyglutamine diseases share a common molecular pathogenesis, namely protein misfolding and aggregation in the brain. Recently, prion-like transmission of these protein aggregates between cells has been suggested to contribute to the spread of neuropathology. To cope with protein aggregation, organisms have protein quality control systems, consisting of molecular chaperones and protein degradation, and therefore, an imbalance between protein aggregation and these systems would lead to neurodegeneration. Mouse models of neurodegenerative diseases exhibit neurological phenotypes before neuronal death, and these phenotypes are unexpectedly reversible, suggesting that neuronal dysfunction rather than death leads to neurological phenotypes. In addition to dysfunctions inside neurons, dysfunctions of the neuronal and neuro-glial networks also contribute to the pathogenesis in a non-cell autonomous fashion. On the other hand, another class of diseases including several spinocerebellar ataxias has been discovered to be caused by a repeat expansion mutation in untranslated RNA resulting in its accumulation, which is called the RNA repeat diseases. Recent discoveries of RNA-binding protein mutations and a repeat expansion mutation in ALS have highlighted the involvement of abnormal RNA metabolism in its pathogenesis. Recently, various researches on molecular-targeted therapies are in progress, which include high throughput chemical screening, RNAi, and gene therapy, etc. Towards development of a therapy for neurodegenerative diseases, sensitive biomarkers suitable for evaluation of therapeutic efficacy in clinical trials are eagerly anticipated.